Agniva Ghosh

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Education

Bachelor of Technology in Computer Science 2024 – 2028
Jalpaiguri Government Engineering College, India

Skills

Languages/Data: Python, C++, SQL, JavaScript, Dart, Pandas, NumPy, Exploratory Data Analysis, Matplotlib
ML & Deep Learning: Scikit-learn, XGBoost, PyTorch, CNNs, LSTMs, Transformers, GNNs, Diffusion, MediaPipe, LoRA
Backend & Deployment: FastAPI, Node.js, Express.js, REST APIs, Git, Vercel, Render
Core CS: Data Structures & Algorithms, OOP, Computer Architecture, Digital Electronics, IoT Architecture

Experience

Research Internship (IIT Kharagpur) June 2026 – Present

- **Developed "LabSync",** an end-to-end IoT laboratory access control and dual-biometric management system using a Node.js Express backend and a cross-platform Flutter dashboard.
- **Architected a dual-auth biometric engine** combining optical fingerprint scanning (ESP32 via UART) with facial recognition (`@vladmandic/face-api`, MobileNet v1), achieving sub-50ms inference.
- **Built a scalable, zero-cost persistent storage layer** using the Google Sheets API for real-time telemetry, access logging, and inventory tracking.
- **Engineered C++ microcontroller firmware** for ESP32 and ESP32-CAM to handle hardware triggers, HTTP polling loops, and live VGA (640x480) camera streaming.

Freelance Machine Learning Engineer 2024 – Present

- Developed an automated Gmail spam filtering and auto-response system for a dairy farm business, processing and storing 100,000+ customer orders in a structured database.
- Built data pipelines for email classification, automated replies, and order data storage, improving operational efficiency and response time.
- Performed detailed exploratory data analysis on vehicle datasets, including data cleaning, feature engineering, and visualization to extract business insights.
- Created analytical reports and dashboards to identify trends, anomalies, and performance metrics from large datasets.

Projects

Telecom Customer Churn & Revenue Intelligence Python, XGBoost, Scikit-learn, FastAPI

- Architected an enterprise-grade scikit-learn pipeline for leak-free data cleaning, categorical preprocessing, and churn prediction using a calibrated **XGBoost Classifier**.
- Integrated **TreeSHAP** to extract top predictive drivers per customer and developed a CLV Action Matrix to dynamically assign personalized retention strategies.
- Implemented real-time data drift monitoring computing Population Stability Index (PSI) and Kolmogorov-Smirnov (KS) statistics against baseline distributions.
- Exposed machine learning inference and drift evaluation endpoints via a robust **FastAPI** service and a Streamlit dashboard.

OncoGraph AI — Clinical Oncology Decision Support orkspace FastAPI, Scikit-learn, NetworkX, NumPy

- Engineered a relational Knowledge Graph using NetworkX mapping 50+ biomarker toxicity nodes, guaranteeing 100% NCCN WHO clinical guideline compliance.
- Designed a collaborative Multi-Agent Swarm orchestrator to simulate multidisciplinary tumor board consensus and active clinical trial matching.
- Implemented prognosis staging prediction via Random Forest models with local Gini importances, DICOM pathology image routing, and FedAvg federated learning.

Sign0 — AI Sign Language Recognition & Synthesizer PyTorch, ONNX, FastAPI, MediaPipe, DDPM Diffusion

- Engineered a real-time Spatial-Temporal Transformer and ONNX MLP pipeline for ASL gesture recognition via webcam, achieving **0.81 ms latency** and **100% validation accuracy**.
- Built a Denoising Diffusion Probabilistic Model (DDPM) text-to-sign synthesizer to generate 3D skeleton gesture animations from text prompts.
- Developed a fully accessible, glassmorphism-styled web platform with an interactive ASL visual dictionary and TTS audio engine.
- Deployed backend models via Render and frontend on Vercel, integrating seamless fallback APIs for continuous availability.

RL Train Football Engine — Multi-Agent Deep RL Simulation PyTorch, Pygame, GNNs, PPO, GANs

- Developed a 100% autonomous 11v11 football simulation leveraging Multi-Head Graph Attention Networks (GAT) to dynamically compute spatial passing channels and defender pressure.
- Trained 22 independent agents using a **Proximal Policy Optimization (PPO)** Actor-Critic neural policy integrated with Generalized Advantage Estimation (GAE).
- Implemented a 1D Convolutional **GAN Trajectory Discriminator** to evaluate physical momentum, heavily penalizing robotic jitter and rewarding fluid, human-like motion.
- Built a 3D aerodynamic physics engine integrating Magnus spin curving, elevation bounds, and collision damping.

Achievements

- Published a research paper in the Regional Science and Technology Congress (2026) on "Solving First-Order Differential Equations using Physics-Informed Neural Networks".
- Built 15+ machine learning projects across NLP, Computer Vision, Multimodal AI, and Time Series.
- Solved 300+ Data Structures and Competitive Programming problems on LeetCode, Codeforces, and Striver SDE Sheet.
- Won prizes in 3 Hackathons, including Google Cloud Hackathons, developing end-to-end AI solutions.